Chapter 7: Sums of Random Variables and Long-Term Averages

7.1 Sums of Random Variables

7.1 $\mathcal{E}[X+Y+Z] = \mathcal{E}[X] + \mathcal{E}[Y] + \mathcal{E}[Z] = 0$

a) From Eqn. 5.3 we have

$$\begin{aligned} VAR(X+Y+Z) &= VAR(X) + VAR(Y) + VAR(Z) \\ &\quad +2COV(X,Y) + 2COV(X,Z) + 2COV(Y,Z) \\ &= 1+1+1+2\left(\frac{1}{4}\right) + 2(0) + 2\left(-\frac{1}{4}\right) = 3 \end{aligned}$$

b) From Eqn. 5.3 we have

$$VAR(X+Y+Z) = VAR(X) + VAR(Y) + VAR(Z) = 3$$

7.2 $\mathcal{E}[S_n] = \mathcal{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathcal{E}[X_i] = n\mu$.

$$VAR(S_n) = \underbrace{\sum_{k=1}^{n} VAR(X_k)}_{\substack{\text{sum of diag.} \\ \text{elements of} \\ \text{covariance matrix } K}} + \underbrace{\sum_{j=1}^{n}\sum_{k=1 j\neq k}^{n} COV(X_j, X_k)}_{\substack{\text{sum of off-diag.} \\ \text{element of } K}}$$

$$K = \begin{bmatrix} \sigma^2 & \rho\sigma^2 & 0 & & \cdots & 0 \\ \rho\sigma^2 & \sigma^2 & \rho\sigma^2 & 0 & \cdots & 0 \\ & & & & & 0 \\ & \ddots & \ddots & \ddots & & \rho\sigma^2 \\ & & & \rho\sigma^2 & & \sigma^2 \end{bmatrix}$$

$\therefore VAR(S_n) = n\sigma^2 + 2(n-1)\rho\sigma^2$

7.3

Proceeding as in previous problem:

$$\begin{aligned}
\mathcal{E}[S_n] &= n\mu \\
K &= \begin{bmatrix} \sigma^2 & \rho\sigma^2 & \rho^2\sigma^2 & \dots & \rho^{n-1}\sigma^2 \\ \rho\sigma^2 & \sigma^2 & \rho\sigma^2 & \dots & \rho^{n-2}\sigma^2 \\ \vdots & & & & \\ \rho^{n-1}\sigma^2 & & \dots & & \sigma^2 \end{bmatrix} \\
VAR(S_n) &= n\sigma^2 + 2\rho\sigma^2 \sum_{j=1}^{n-1}\sum_{k=0}^{j-1} \rho^k \\
&= n\sigma^2 + 2\rho\sigma^2 \sum_{j=1}^{n-1} \frac{1-\rho^j}{1-\rho} \\
&= n\sigma^2 + 2\rho\sigma^2 \left[\frac{n-1}{1-\rho} - \left(\frac{\rho}{1-\rho}\right)\frac{1-\rho^{n-1}}{1-\rho}\right]
\end{aligned}$$

7.4

a) By Eqn. 7.7 we have

$$\Phi_Z(\omega) = \Phi_X(\omega)\Phi_Y(\omega) = e^{-\alpha|\omega|}e^{-\beta|\omega|} = e^{-(\alpha+\beta)|\omega|}$$

b) Taking the inverse transform:

$$f_Z(z) = \Phi_Z^{-1}(\omega) = \frac{1}{\lambda}\frac{\alpha+\beta}{(\alpha+\beta)^2+z^2} \Rightarrow Z$$

is also Cauchy

7.5

$\Phi_{S_k}(\omega) = \left(\frac{1}{1-2j\omega}\right)^{\frac{n_1}{2}}\left(\frac{1}{1-2j\omega}\right)^{\frac{n_2}{2}} \dots \left(\frac{1}{1-2j\omega}\right)^{\frac{n_k}{2}} = \left(\frac{1}{1-2j\omega}\right)^{\frac{n_1+n_2+\dots+n_k}{2}}$
$\Rightarrow S_k$ is chi-square RV with $n = n_1 + n_2 + \dots + n_k$.

7.6

a) From Ex.4.34 X_i^2 is chi-square with one degree of freedom. From Prob.7.5, S_n is then chi-square with n degrees of freedom

b)
$$\begin{aligned} T_n &= \sqrt{S_n} \\ \Rightarrow f_{T_n}(x) &= \frac{f_{S_n}(x^2)}{\frac{1}{2}|(x^2)^{-\frac{1}{2}}|} = 2x f_{X_n}(x^2) \end{aligned}$$

Now use fact that S_n is chi-square:

$$= \frac{2x(x^2)^{\frac{n-2}{2}} e^{-x^2/2}}{2^{n/2}\Gamma(n/2)} = \frac{x^{n-1}e^{-x^2/2}}{2^{n/2-1}\Gamma\left(\frac{n}{2}\right)} \qquad x > 0$$

c) $f_{T_2}(x) = xe^{-x^2/2}\ x > 0$

d) $f_{T_3}(x) = \dfrac{x^2e^{-x^2/2}}{2^{1/2}\Gamma\left(\frac{3}{2}\right)} = \dfrac{x^2e^{-x^2/2}}{\sqrt{2}\frac{1}{2}\Gamma\left(\frac{1}{2}\right)} = \sqrt{\dfrac{2}{\pi}}x^2e^{-x^2/2}\ x > 0$

7.7

a) $\Phi_Z(\omega) = \left(\dfrac{\alpha}{\alpha - j\omega}\right)\left(\dfrac{\beta}{\beta - j\omega}\right)$

b) $\Phi_Z(\omega) = \dfrac{a}{\alpha - j\omega} + \dfrac{b}{\beta - j\omega}$ is partial fraction expansion, where $b = \dfrac{\alpha - \beta}{\alpha\beta}$, $a = \dfrac{\beta - \alpha}{\alpha\beta}$

$$\begin{aligned} \Phi_Z(\omega) &= \frac{\beta - \alpha}{\alpha^2\beta}\left(\frac{\alpha}{\alpha - j\omega}\right) - \frac{\beta - \alpha}{\alpha\beta^2}\left(\frac{\beta}{\beta - j\omega}\right) \\ \Rightarrow f_Z(t) &= \mathcal{F}^{-1}[\Phi_Z(\omega)] = \frac{\beta - \alpha}{\alpha\beta}e^{-\alpha t} - \frac{\beta - \alpha}{\alpha\beta}e^{-\beta t} \qquad t > 0 \end{aligned}$$

7.8

a) $\Phi_Z 9\omega) = \mathcal{E}[e^{j(aX+bY)}] = \mathcal{E}[e^{jaX}]\mathcal{E}[e^{-jbY}] = \Phi_X(a\omega)\Phi_Y(b\omega)$

b)
$$\begin{aligned} \mathcal{E}[Z] &= \frac{1}{j}\Phi'_Z(\omega)|_{\omega=0} = \frac{1}{j}\Phi'_X(a\omega)a\Phi_Y(b\omega)|_{\omega=0} + \frac{1}{j}\Phi'_Y(b\omega)b\Phi_X(a\omega|_{\omega=0} \\ &= a\mathcal{E}[X] + b\mathcal{E}[Y] \\ \mathcal{E}[Z^2] &= -\Phi''_Z(\omega) \\ &= -[\Phi''_X(a\omega)a^2\Phi_Y(b\omega) + 2\Phi'_X(a\omega)a\Phi'_Y(b\omega)b + \Phi_X(a\omega)\Phi_Y(b\omega)b^2]_{\omega=0} \\ &= a^2\mathcal{E}[X^2] + b^2\mathcal{E}[Y^2] - 2ab\mathcal{E}[X]\mathcal{E}[Y] \\ VAR[Z] &= \mathcal{E}[Z^2] - \mathcal{E}[Z]^2 = a^2VAR[X] + b^2\varepsilon[Y] \end{aligned}$$

7.9

$$\begin{aligned}\Phi_{M_n}(\omega) &= \mathcal{E}[e^{j\omega M_n}] = \mathcal{E}[e^{j\omega \frac{1}{n}\sum_{j=1}^{n} X_j}] \\ &= \prod_{j=1}^{n} \mathcal{E}[e^{j\frac{\omega}{n}X_j}] \\ &= \left(\Phi_X\left(\frac{\omega}{n}\right)\right)^n\end{aligned}$$

7.10

$$\begin{aligned}G_{S_k}(z) &= \mathcal{E}[z^{X_1+\ldots+X_k}] = \mathcal{E}[z^{X_1}]\ldots\mathcal{E}[z^{X_k}] = G_{X_1}(z)\ldots G_{X_k}(z) \\ &= [pz+q]^{n_1}[pz+q]^{n_2}\ldots[pz+q]^{n_k} \\ &= [pz+q]^{n_1+\ldots+n_k}\end{aligned}$$

where the second equality follows from the independence of the X_i's. The result states that S_k is Binomial with parameters $n_1 + \ldots + n_k$ and p. This is obvious since S_k is the number of heads in $n_1 + \ldots + n_k$ tosses.

7.11

$$\begin{aligned}G_{S_k}(z) &= G_{X_1}(z)\ldots G_{X_k}(z) = e^{\alpha_1(z-1)}e^{\alpha_2(z-1)}\ldots e^{\alpha_k(z-1)} \\ &= e^{(\alpha_1+\ldots+\alpha_k)(z-1)}\end{aligned}$$

$\Rightarrow X_k$ Poisson with rate $\alpha_1 + \ldots + \alpha_k$.

7.12

) Note first that

$$\mathcal{E}[S/N = n] = \mathcal{E}\left[\sum_{k=1}^{n} X_k\right] = nE[X] \ ,$$

thus

$$\mathcal{E}[S] = \mathcal{E}[\mathcal{E}[S/N]] = \mathcal{E}[N\mathcal{E}[X]] = \mathcal{E}[N]\mathcal{E}[X] \ .$$

$$\mathcal{E}[S^2] = \mathcal{E}[\mathcal{E}[S^2/N]]$$

which requires that we find

$$\begin{aligned}\mathcal{E}[S^2|N = n] &= \mathcal{E}\left[\sum_{i=1}^{n} X_i \sum_{j=1}^{n} X_j\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathcal{E}[X_i X_j] \\ &= n\mathcal{E}[X^2] + n(n-1)\mathcal{E}[X]^2\end{aligned}$$

since $E[X_i X_j] = \mathcal{E}[X^2]$ if $i = j$ and $E[X_i X_j] = \mathcal{E}[X]^2$ if $i \neq j$. Thus

$$\begin{aligned}\mathcal{E}[S^2] &= \mathcal{E}[N\mathcal{E}[X^2] + N(N-1)\mathcal{E}[X]^2] \\ &= \mathcal{E}[N]\mathcal{E}[X^2] + \mathcal{E}[N^2]\mathcal{E}[X^2] - \mathcal{E}[N]\mathcal{E}[X]^2\end{aligned}$$

Then

$$\begin{aligned}VAR(S) &= \mathcal{E}[S^2] - \mathcal{E}[S]^2 \\ &= \mathcal{E}[N]\mathcal{E}[X^2] + \mathcal{E}[N^2]\mathcal{E}[X^2] - \mathcal{E}[N]\mathcal{E}[X]^2 - \mathcal{E}[N]^2\mathcal{E}[X]^2 \\ &= \mathcal{E}[N]VAR[X] + VAR[N]\mathcal{E}[X]^2\end{aligned}$$

b) First note that

$$\mathcal{E}[z^S/N = n] = \mathcal{E}[z^{\sum_{i=1}^{n} X_i}] = \mathcal{E}[z^{X_1}]...\mathcal{E}[z^{X_n}] = G_X(z)^n$$

Then

$$\begin{aligned}\mathcal{E}[z^S] &= \mathcal{E}[\mathcal{E}[z^S|N] \\ &= \mathcal{E}[G_X^N(z)] \\ &= \mathcal{E}[\omega^N]_{\omega=G_X(z)} \\ &= G_N(G_X(z))\end{aligned}$$

7.13)

$$X_j = \begin{cases} 500 & 1/2 \\ 1000 & 1/2 \end{cases} \qquad E[X] = 750 \;, \; E[X^2] = 625000$$

a) $R = \sum_{j=1}^{N} X_j$, from problem 7.12, we have: $E[R] = E[N]\,E[X]$

So, $E[R] = 750L$,

Also $VAR(R) = E[N]\,VAR[X] + VAR[N]\,E[X]^2$. Since $E[N] = VAR[N] = L$

Then $VAR(R) = L\left[VAR[X] + E^2[X]\right] = L\,E[X^2] = 625000L$

b) $G_R = G_N(G_X(z))$ (P.7.12), $G_X(z) = \frac{1}{2}(z^{500} + z^{1000})$, $G_N(z) = e^{L(z-1)}$

Therefore: $G_R(z) = e^{L(\frac{1}{2}z^{500} + \frac{1}{2}z^{1000} - 1)}$

7.14

N # of widgets tested in 1-hour $N \sim Binomial(600, p)$

$X_i \sim Bernoulli(a)$

$S = \sum_{i=1}^{N} X_i$

a) from P 7.12: $E[S] = E[N]\,E[X] = np \times a = 600pa$

$VAR(S) = E[N]\,VAR[X] + VAR[N]\,E[X]^2 = 600p(1-a)a + 600p(1-p)a^2$

b) from P.7.12

$G_S(z) = G_N(G_X(z))$, $G_N(z) = (1-p+pz)^{600}$, $G_X(z) = (1-a+az)$

Therefore $G_S(z) = (1-p+p(1-a+az))^{600} = (1-pa+paz)^{600}$

7.2 The Sample Mean and the Laws of Large Numbers

7.15

$$P\left[\left|\frac{N(t)}{t} - \lambda\right| \geq \varepsilon\right] = P[|N(t) - \lambda t| \geq \varepsilon t]$$
$$\leq \frac{VAR[N(t)]}{(\varepsilon t)^2} \quad \text{by Chebyshev Inq.}$$
$$= \frac{\lambda t}{\varepsilon^2 t^2} = \frac{\lambda}{\varepsilon^2 t}$$

7.16

$p = \frac{2}{10}$

$$P[|f_A(n) - p| < \varepsilon] \geq 1 - \frac{p(1-p)}{n\varepsilon^2} = 0.95$$

letting $p = \frac{2}{10}$, $\varepsilon = \frac{1}{50} \Rightarrow n = 8000$

7.17

$$M_{20} = \frac{1}{20}(X_1 + \ldots + X_{20}) = \frac{1}{20} S_{20}$$
$$\mu = \mathcal{E}[X] = \frac{1+2+\ldots+6}{6} = 3.5$$
$$\sigma_X^2 = \frac{1}{6}(1 + s^2 + 3^2 + 4^2 + 5^2 + 6^2) - (3.5)^2 = 2.91667$$

$$P[60 < S_{100} < 80] = \left[3 < \frac{S_{100}}{20} \leq 4\right]$$
$$= P[-.5 < M_{100}^{20} - 3.5 < .5]$$
$$= P[|M_{100}^{20} - 3.5| < .5]$$
$$\geq 1 - \frac{2.92}{20(.5)^2} = 0.416$$

7.18 For $n = 16$, Eqn. 7.20 gives

$$P[|M_{16} - 0| < \varepsilon] \geq 1 - \frac{1^2}{16\varepsilon^2} = 1 - \frac{1}{16}\frac{1}{\varepsilon^2}$$

Since M_{16} is Gaussian with mean 0 and variance $\frac{1}{16}$

$$\begin{aligned} P[|M_{16} - 0| < \varepsilon] &= P[-\varepsilon < M_{16} \leq \varepsilon] = 1 - 2Q(\sqrt{16}\varepsilon) \\ &= 1 - 2Q(3.16\varepsilon) \end{aligned}$$

Similarly for $n = 100$ we obtain

$$\begin{aligned} P[|M_{100} - 0] < \varepsilon] &\geq 1 - \frac{1}{100}\frac{1}{\varepsilon^2} \\ P[|M_{100} - 0] < \varepsilon] &= 1 - 2Q(10\varepsilon) \end{aligned}$$

For example if $\varepsilon = \frac{1}{2}$

$$\begin{aligned} P[|M_{16}| < \tfrac{1}{2}] &\geq 1 - \frac{1}{16/4} = .75 \\ P[|M_{16}| < \tfrac{1}{2}] &= 1 - 2Q(2.5) = 1 - 2(5.44(10^{-2})) = .890 \\ P[|M_{100}| < \tfrac{1}{2}] &\geq 1 - \frac{1}{100/4} = .96 \\ P[|M_{100}| < \tfrac{1}{2}] &= 1 - 2Q(5) = 1 - 2(2.87)10^{-6} \end{aligned}$$

Note the significant discrepancies between the bounds and the exact values.

7.19

$$\begin{aligned} P\left[\left|\frac{1}{n}S_n - \mu\right| > \varepsilon\right] &\leq \frac{VAR(\frac{1}{n}S_n)}{\varepsilon^2} = \frac{VAR(S_n)}{n^2\varepsilon^2} \\ &= \frac{n\sigma^2 + 2(n-1)\rho\sigma^2}{n^2\varepsilon^2} \to 0 \quad \text{as } n \to \infty \end{aligned}$$

$\Rightarrow$ Week Law of Large Numbers holds.

7.20

$$\begin{aligned} P\left[\left|\frac{1}{n}S_n - \mu\right| > \varepsilon\right] &\leq \frac{VAR(S_n)}{n^2\varepsilon^2} \\ &= \frac{1}{n^2\varepsilon^2}\left[n\sigma^2 + 2\rho\sigma^2\left(\frac{n-1}{1-\rho} - \frac{\rho}{1-\rho}\frac{1-\rho^{n-1}}{1-\rho}\right)\right] \\ &= \frac{\sigma^2}{n\varepsilon^2} + \frac{2\rho\sigma^2}{\varepsilon^2}\left(\frac{n-\frac{1}{n}}{n(1-\rho)} - \frac{1}{n^2}\frac{\rho(1-\rho^{n-1})}{(1-\rho)^2}\right) \\ &\to 0 \text{ as } n \to \infty \text{ (assuming } \rho < 1) \end{aligned}$$

$\Rightarrow$ Weak Law of Large Numbers holds.

7.21

a)
$$\begin{aligned} LHS &= \sum_{j=1}^{n}(X_j^2 - 2\mu X_j + \mu^2) = \sum_{j=1}^{n} X_j^2 - 2\mu(nM_n) + n\mu^2 \\ RHS &= \sum_{j=1}^{n}(X_J^2 - 2M_nX_j + M_n^2) + n(M_n - \mu)^2 \\ &= \sum_{j=1}^{n} X_j^2 - 2M_n(nM_n) + nM_n^2 + nM_n^2 - 2n\mu M_n + n\mu^2 \\ &= \sum_{j=1}^{n} X_j^2 - 2n\mu M_n + n\mu^2 = LHS \quad \surd \end{aligned}$$

b)
$$\begin{aligned} \mathcal{E}\left[k\sum_{j=1}^{n}(X_j - M_n)^2\right] &= k\mathcal{E}\left[\underbrace{\sum_{j=1}^{n}(X_j - \mu)^2 - n(M_n - \mu)^2}_{\text{from part a}}\right] \\ &= k\sum_{j=1}^{n}\mathcal{E}[(X_j - \mu)^2] - kn\mathcal{E}[(M_n - \mu)^2] \\ &= kn\sigma^2 - kn\frac{\sigma^2}{n} \\ &= k(n-1)\sigma^2 \quad \text{since } VAR[M_n] = \tfrac{\sigma^2}{n}. \end{aligned}$$

c) If $k = \frac{1}{n-1}$ then $\mathcal{E}[V_n^2] = \sigma^2$

d) if $k = \frac{1}{n}$ then

$$\mathcal{E}\left[\frac{1}{n}\sum_{j=1}^{n}(X_j - M_n)^2\right] = \left(1 - \frac{1}{n}\right)\sigma^2 = \sigma^2 - \underbrace{\frac{1}{n}\sigma^2}_{\text{bias}}$$

7.3 The Central Limit Theorm

7.22

The relevant parameters are $n = 100$, $m = np = 50$, $\sigma^2 = npq = 25$. The Central Limit Theorem then gives:

$$P[40 \leq N \leq 60] = P\left[\frac{40-50}{\sqrt{25}} \leq \frac{N-m}{\sigma} \leq \frac{60-50}{\sqrt{25}}\right]$$

-2 $\qquad$ 2

$$\approx Q(-6.324) - Q(6.324) = 1 - 2Q(6.324) = 1 - 2.54(10^{-10}) \quad 2(2.28)\times 10^{-2} = 0.9544$$

$$P[50 \leq N \leq 55] \approx Q(0) - Q(3.162) = \frac{1}{2} - \underbrace{7.3(10^{-4})}_{.159} = 0.441$$

7.23

The relative frequency $f_A(n)$ has mean $\frac{2}{10}$ and variance $\frac{1}{n}p(1-p) = \frac{0.16}{n}$

$$P[|f_Z(n) - 0.2| < 0.02] = P[0.18 < f_A(n) < 0.22]$$

$$= P\left[\frac{0.18-0.20}{\sqrt{\frac{0.16}{n}}} < \frac{f_A(n)-0.1}{\sqrt{\frac{0.16}{n}}} < \frac{0.22-0.20}{\sqrt{\frac{0.16}{n}}}\right]$$

$$\approx 1 - 2Q\left(\frac{0.02}{\sqrt{\frac{0.16}{n}}}\right) = 0.95$$

$$\Rightarrow Q\left(\frac{\sqrt{n}}{20}\right) = 0.025 \Rightarrow \frac{\sqrt{n}}{20} = 1.95 \Rightarrow n = 1521$$

7.24

$S = \sum_{i=1}^{20} X_i \Rightarrow E[S] = 20 \cdot E[X] = 20 \times 3.5 = 70$

$VAR[S] = 20\, VAR[X] = 20 \times 2.92 = 58.4$

using CLT we have: $S \sim N(70, \sqrt{58.4})$

$$P\{60 < S < 80\} = P\left\{\frac{60-70}{7.64} < \frac{S-70}{7.64} < \frac{80-70}{7.64}\right\}$$

$$= 1 - 2Q(1.3089) = 0.8094$$

7.25

$\mathcal{E}[X_i] = \frac{1}{\lambda} = 36 \qquad VAR(X_i) = \frac{1}{\lambda^2} = 36^2$

$$S = X_1 + ... + X_{16} \qquad \mathcal{E}[S] = 16(36) \qquad VAR(S) = 16(36)^2$$

$$\begin{aligned} P[S < 600] &= P\left[\frac{S - 16(36)}{4(36)} < \frac{600 - 16(36)}{4(36)}\right] \\ &\cong 1 - Q\left(\frac{1}{6}\right) = 0.5692 \end{aligned}$$

7.26

$$\begin{aligned} \mathcal{E}[S_n] &= n\mathcal{E}[X_i] = n \cdot 1 = n \\ VAR[S_n] &= n\sigma_{x_i}^2 = n \cdot 1^2 = n \end{aligned}$$

Assuming S_n approximately Gaussian:

$$P[S_n > 15] = P\left[\frac{S_n - n}{\sqrt{n}} > \frac{15 - n}{\sqrt{n}}\right] \approx Q\left(\frac{15 - n}{\sqrt{n}}\right) = 0.99$$

From Table 3.4

$$\frac{15 - n}{\sqrt{n}} = -2.3263$$

$$\Rightarrow n - 2.3263\sqrt{n} - 15 = 0 \Rightarrow n = 27.04$$

$\Rightarrow$ by 28 pens

7.27 $n = 80 \qquad \lambda = \frac{1}{4}$

$$\mathcal{E}[S_n] = n\lambda = 20 \qquad VAR(S_n) = 20$$

$$\begin{aligned} P[S_n = k] &\approx \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} \frac{e^{-(x-20)^2/2(20)}}{\sqrt{2\pi(20)}} dx \quad \text{as per Eqn. 7.29} \\ &\approx \frac{e^{-(k-20)^2/40}}{\sqrt{40\pi}} \quad \text{as per Eqn. 7.30} \end{aligned}$$

A comparison of the exact value of $P[S_n = k]$ and the above approximation is given below:

k	poisson	approx.	k	poisson	approx
0	0.000000	0.000004	21	0.084605	0.087003
1	0.000000	0.000010	22	0.076913	0.080717
3	0.000002	0.000064	23	0.066881	0.071232
4	0.000013	0.000148	24	0.055734	0.059796
5	0.000054	0.000321	25	0.044587	0.047748
6	0.000183	0.000664	26	0.034298	0.036268
7	0.000523	0.001304	27	0.025406	0.026205
8	0.001308	0.002437	28	0.018147	0.018010
9	0.002908	0.004331	29	0.012515	0.011774
10	0.005816	0.007322	30	0.008343	0.007322
11	0.010575	0.011774	31	0.005382	0.004331
12	0.017625	0.018010	32	0.003364	0.002437
13	0.027115	0.026205	33	0.002038	0.001304
14	0.038736	0.036268	34	0.001199	0.000664
15	0.051648	0.047748	35	0.000685	0.000321
16	0.064561	0.059796	36	0.000380	0.000148
17	0.075954	0.071232	37	0.000205	0.000064
18	0.084393	0.080717	38	0.000108	0.000027
19	0.088835	0.087003	39	0.000055	0.000010
20	0.088835	0.089206	40	0.000027	0.000004

7.28

5.28 Using the fact that the time between arrivals is an exponential RV with mean $\frac{1}{15}$ and variance $\left(\frac{1}{15}\right)^2$; the time of the nth arrival is:

$$S_n = X_1 + ... + X_n$$

where

$$\begin{aligned} \mathcal{E}[S_n] &= n\mathcal{E}[X_i] = n/15 \\ VAR[S_n] &= nVAR[X_i] = n/225 \end{aligned}$$

$$\begin{aligned} P\left[\begin{array}{c}\text{more than 950}\\ \text{messages in}\\ \text{60 seconds}\end{array}\right] &= P[S_{950} < 60] \\ &= P\left[\frac{S_{950} - 63.33}{\sqrt{4.22}} < \frac{60 - 63.33}{\sqrt{4.22}}\right] \\ &\approx 1 - Q(-1.96) = Q(1.96) = 2.49(10^{-2}) \\ &\approx Q(-6.324) - Q(6.324) = 1 - 2Q(6.324) \\ &= 1 - 2.54(10^{-10}) \end{aligned}$$

$$P[50 \leq N \leq 55] \approx Q(0) - Q(3.162) = \frac{1}{2} - 7.3(10^{-4})$$

$E[S_{950}] = 63.33$

$VAR[S_{950}] = 4.22$

5.4

-1.62 1.62

-2 2 2

$2(2.28)\times10^{-2} = 0.9544$

1 $.159$

$= 0.441$

7.29

5.29 The total number of errors S_{100} is the sum of iid Bernoulli random variables

$$\begin{aligned} S_{100} &= X_1 + ... + X_{100} \\ \mathcal{E}[S_{100}] &= 100p = 15 \\ VAR[S_{100}] &= 100pq = 12.75 \end{aligned}$$

The Central Limit Theorem gives:

$$\begin{aligned} P[S_{100} \leq 20] &= 1 - P[S_{100} > 20] \\ &= 1 - P\left[\frac{S_{100} - 15}{\sqrt{12.75}} > \frac{20 - 15}{\sqrt{12.75}}\right] \\ &\approx 1 - Q(1.4) = 0.92 \end{aligned}$$

7.30

5.30 Total error is

$$S_{64} = X_1 + X_2 + ... + X_{64}$$

where X_i uniform is $[-\frac{1}{2}, \frac{1}{2}]$

$$\mathcal{E}[X_i] = 0 \qquad VAR[X_i] = \frac{1}{12}$$

$$P[S_{64} > 4] = P\left[\frac{S_{100}}{\sqrt{\frac{64}{12}}} > \frac{4}{\sqrt{\frac{64}{12}}}\right] \approx Q(2.078) = 1.79(10^{-2})$$

4.16

1.7321

7.31

1) $X_i \sim$ Bernoulli $(1/2)$ $\quad X_i = \begin{cases} 1 & \text{head} \\ 0 & \text{tail} \end{cases}$

$S_{100} = \sum_{i=1}^{100} X_i \quad E[S_{100}] = 50, \quad VAR[S_{100}] = 25$

a) $P\{S_{100} \geq 91\} \leq \left(\frac{\frac{1}{2}}{\left(\frac{91}{100}\right)^{\frac{91}{100}} \times \left(\frac{9}{100}\right)^{\frac{9}{100}}}\right)^{100} = 1.0886 \times 10^{-17}$

using CLT: $P\{S_{100} \geq 91\} = Q\left(\frac{91-50}{5}\right) = Q(8.2) = 1.11 \times 10^{-16}$

b) $P\{S_{1000} \geq 651\} \leq \left(\frac{\frac{1}{2}}{\left(\frac{651}{1000}\right)^{\frac{651}{1000}} \left(\frac{349}{1000}\right)^{\frac{349}{1000}}}\right)^{1000} = 7.6332 \times 10^{-21}$

using CLT: $P\{S_{1000} > 651\} = Q\left(\frac{651-500}{\sqrt{250}}\right) = Q(9.5501) = 6.47 \times 10^{-22}$

7.32)

Total error $E_{100} = \sum_{i=1}^{100} X_i$, X_i represents an error in ith transmission

$$P\{E_{100} \geq 4\} \leq \left(\frac{0.01^{0.04} \times 0.99^{0.96}}{0.01^{0.04} \times 0.96^{0.96}} \right)^{100} = 0.0749$$

using CLT : $P\{E_{100} \geq 4\} \approx Q\left(\frac{4-1}{\sqrt{0.99}}\right) = Q(3.0151) = 0.0013$

7.33

$$W_{20} = \sum_{i=1}^{20} X_i \qquad X_i = \begin{cases} 1 & p = \frac{2}{5} \\ 0 & p = 3/5 \end{cases}$$

a) $$P\{W_{20} \geq 11\} \leq \left(\frac{(\frac{2}{5})^{11/20} \times (\frac{3}{5})^{9/20}}{(\frac{11}{20})^{11/20} \times (\frac{9}{20})^{9/20}} \right)^{20} = 0.4010$$

b) $$P\{W_{100} \geq 51\} \leq \left(\frac{(\frac{2}{5})^{\frac{51}{100}} \times (\frac{3}{5})^{\frac{49}{100}}}{(\frac{51}{100})^{0.51} \times (\frac{49}{100})^{0.49}} \right)^{100} = 0.0849$$

c) $$P\{W_n \geq \frac{n}{2}+1\} \leq \left(\frac{(\frac{2}{5})^{\frac{1}{2}+\frac{1}{n}} \times (\frac{3}{5})^{\frac{1}{2}-\frac{1}{n}}}{(\frac{1}{2}+\frac{1}{n})^{(\frac{1}{2}+\frac{1}{n})} (\frac{1}{2}-\frac{1}{n})^{(\frac{1}{2}-\frac{1}{n})}} \right)^{n} = P$$

for $n = 92$, $P = 0.0998$

7.34

$$P[X \geq a] \leq e^{-sa} E[e^{sX}] \quad , s>0$$

$$G_X(z) = e^{\alpha(z-1)}$$

$$E[(e^s)^X] = e^{\alpha(e^s-1)}$$

therefore $P[X \geq a] \leq e^{-sa + \alpha e^s - \alpha}$, $s>0$

To minimize bound we have to find the root of power.

So: $f(s) = -sa + \alpha e^s - \alpha$, $f'(s) = -a + \alpha e^s = 0 \Rightarrow e^s = \frac{a}{\alpha} \Rightarrow s = \ln \frac{a}{\alpha}$

if $\ln \frac{a}{\alpha} \geq 0$ or $a \geq \alpha$

Therefore: $P[X \geq a] \leq e^{-a \ln \frac{a}{\alpha} + a - \alpha}$

7.35

number of faulty pens in the duration of 15 weeks is Poisson RV with mean 15. So: $S \sim$ Poisson (15)

according to P 7.34 we have

$$P[S > a] \leq e^{-a \ln(\frac{a}{15}) + a - 15} \quad \text{for } a > 15$$

Therefore: $e^{-a \ln(\frac{a}{15}) + a - 15} = 0.01$

$$\Rightarrow -a \ln(\frac{a}{15}) + a - 15 = \ln 0.01$$

by trial & error we find: $a = 28$

So, the student should buy 28 pens.

7.36)

$$P[X \geq a] \leq e^{-sa} E[e^{sX}] \quad , s>0$$

$X \sim N(\mu, b)$

$$E[e^{sX}] = e^{\mu s + \frac{\delta^2 s^2}{2}}$$

Therefore $P[X \geq a] \leq e^{-sa + \mu s + \frac{\delta^2 s^2}{2}}$

$$f(s) = -sa + \mu s + \frac{\delta^2 s^2}{2} \Longrightarrow f'(s) = -a + \mu + \delta^2 s = 0 \Rightarrow s^* = \frac{a-\mu}{\delta^2} \quad \text{if } a > \mu$$

$$\Rightarrow P[X \geq a] \leq e^{-a\left(\frac{a-\mu}{\delta^2}\right) + \mu\left(\frac{a-\mu}{\delta^2}\right) + \frac{1}{2\delta^2}(a-\mu)^2}$$

$$= e^{-\frac{(a-\mu)^2}{2\delta^2}} \quad , a > \mu$$

7.37)

$$P[X \geq a] = Q\left(\frac{a-\mu}{\delta}\right) \approx \frac{1}{\left(1-\frac{1}{\pi}\right)\left(\frac{a-\mu}{\delta}\right) + \frac{1}{\pi}\sqrt{\left(\frac{a-\mu}{\delta}\right)^2 + 2\pi}} e^{-\frac{(a-\mu)^2}{2\delta^2}} \quad \text{for } a > \mu$$

if you compare this value with the bound in (P7.36) you can see that the difference is coefficient $\dfrac{1}{\left(1-\frac{1}{\pi}\right)\left(\frac{a-\mu}{\delta}\right) + \frac{1}{\pi}\sqrt{\left(\frac{a-\mu}{\delta}\right)^2 + 2\pi}}$

7.38

a) $P[X \geq a] \leq e^{-sa} E[e^{sX}]_{,s>0}$ $X \sim \exp(\lambda)$, $E[e^{sX}] = \frac{\lambda}{\lambda - s}$

Therefore:

$$P[X \geq a] \leq e^{-sa} \times \frac{\lambda}{\lambda - s}$$

$$f(s) = e^{-sa} \times \frac{\lambda}{\lambda - s}, \quad f'(s) = \frac{-ae^{-sa}\lambda(\lambda - s) + \lambda e^{-sa}}{(\lambda - s)^2}$$

$f'(s) = 0 \Rightarrow s^* = \lambda - \frac{1}{a}$, since $s^* > 0$ then $\lambda > 1/a$ or $a > 1/\lambda$

Therefore $P[X \geq a] \leq a\lambda e^{-(\lambda a - 1)}$ for $a > 1/\lambda$

b) we know $P[X \geq \frac{k}{\lambda}] = e^{-\lambda \times \frac{k}{\lambda}} = e^{-k}$

if $a = \frac{k}{\lambda}$, chernoff's bound gives us: $P[X \geq \frac{k}{\lambda}] \leq ke^{-(k-1)}$, $k > 1$

$ke^{-k+1} = ke\, e^{-k}$, Therefore chernoff bound is always greater than actual probability, with a factor of ke

7.39 $X \sim \Gamma(\lambda, \alpha)$, $E[e^{sX}] = \dfrac{1}{(1-\frac{s}{\lambda})^{\alpha}}$

a)

$$P[X \geq a] \leq e^{-sa} \times \frac{1}{(1-\frac{s}{\lambda})^{\alpha}}$$

$$f(s) = e^{-sa} \times \frac{1}{(1-\frac{s}{\lambda})^{\alpha}} \quad , \quad f'(s) = \frac{-a e^{-sa}(1-\frac{s}{\lambda})^{\alpha} + e^{-sa}\alpha\frac{1}{\lambda}(1-\frac{s}{\lambda})^{\alpha-1}}{(1-\frac{s}{\lambda})^{2\alpha}}$$

$$f'(s) = 0 \Longrightarrow s^* = \lambda - \frac{\alpha}{a} \text{ , since } s>0 \Rightarrow a > \frac{\alpha}{\lambda}$$

Therefore: $$P[X \geq a] \leq e^{-(\lambda a - \alpha)} \frac{\lambda^{\alpha}}{(a\alpha)^{\alpha}} \quad \text{for } a > \frac{\alpha}{\lambda}$$

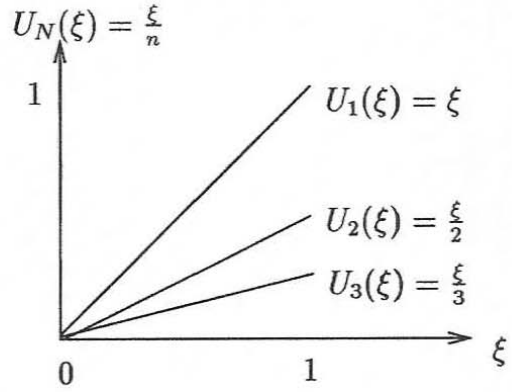
b) chi-square with k degree of freedom, $X \sim \text{chi-square}(k) \sim \Gamma(\frac{1}{2}, \frac{k}{2})$

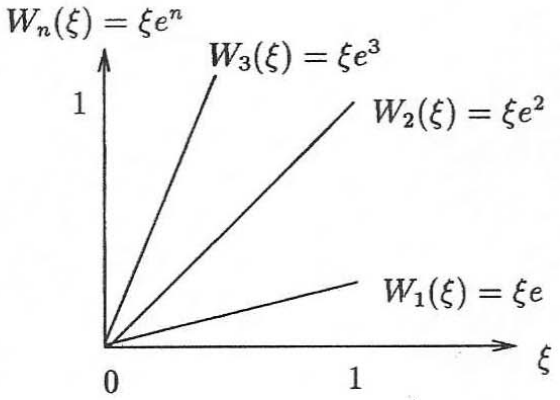
$\alpha = \frac{k}{2}$, $\lambda = \frac{1}{2}$ from part a: $P[X \geq a] \leq e^{-\frac{1}{2}(a-k)} (ak)^{-k/2}$ for $a > k$

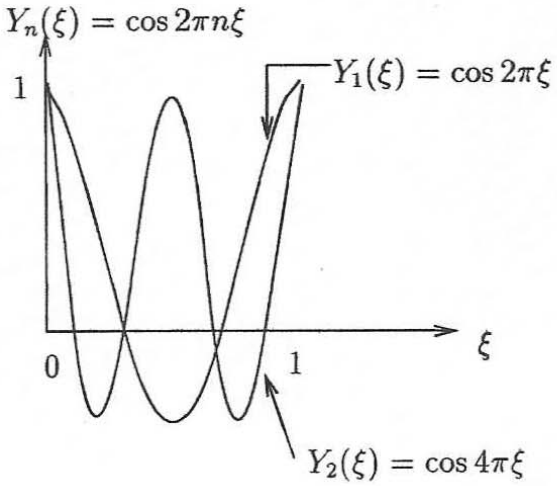
*7.4 Convergence of Sequences of Random Variables

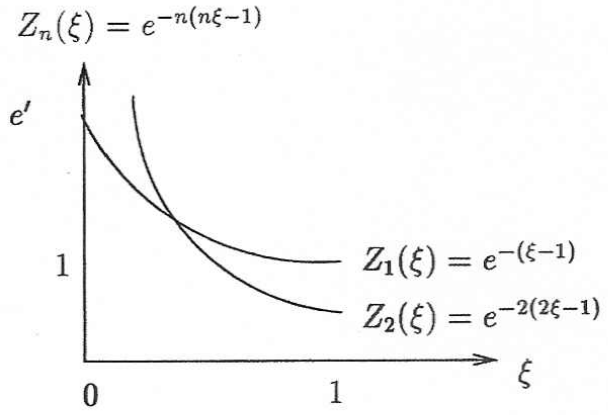
7.40

5.40 a)

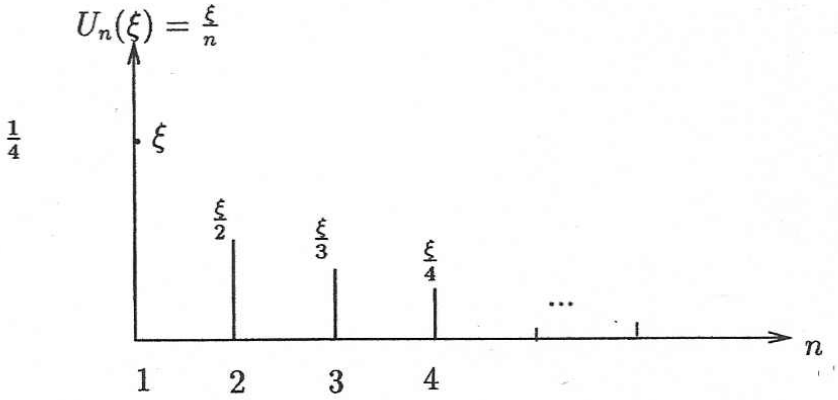


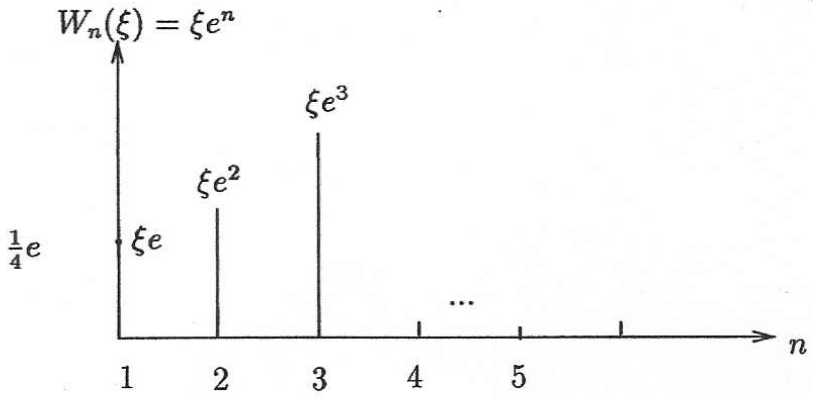


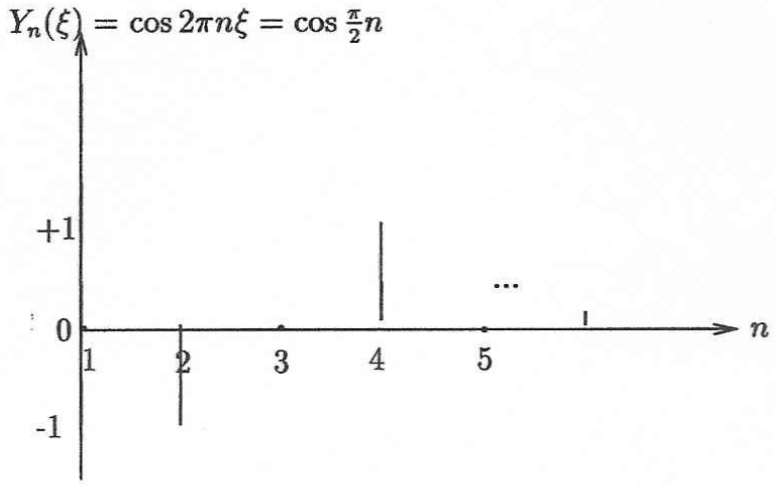


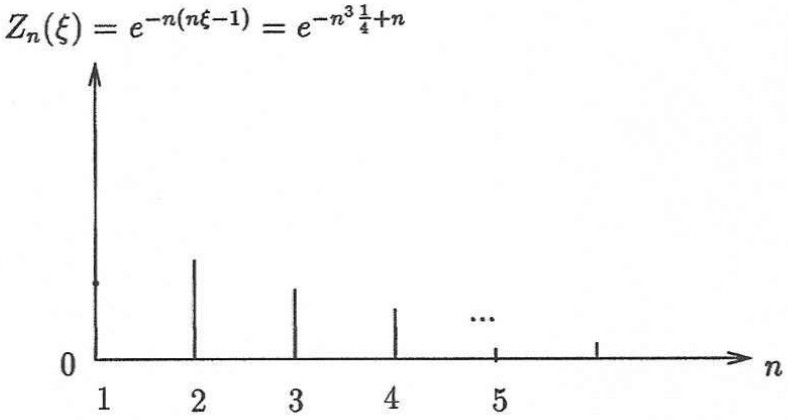

$Z_n(\xi) = e^{-n(n\xi-1)}$
e'
1
$Z_1(\xi) = e^{-(\xi-1)}$
$Z_2(\xi) = e^{-2(2\xi-1)}$
ξ
0
1

b)


$U_n(\xi) = \frac{\xi}{n}$
$\frac{1}{4}$
ξ
$\frac{\xi}{2}$
$\frac{\xi}{3}$
$\frac{\xi}{4}$
...
n
1
2
3
4


$W_n(\xi) = \xi e^n$
ξe^3
ξe^2
$\frac{1}{4}e$
ξe
...
n
1
2
3
4
5





7.41

$X_n(\xi) = \xi^n \to 0$ for all $0 \leq \xi < 1$

$X_n(\xi)$ converges to 0 almost surely

$Y_n(\xi) = \cos^2 2\pi\xi$ (seems to be $\cos^2 2\pi n\xi$)

$\cos^2 2\pi\xi$ doesn't depend on n

$Y_n(\xi) = \cos^2 2\pi\xi \to Y(\xi) = \cos^2 2\pi\xi$ for all $\xi \in S$

$Z_n(\xi) = \cos^n 2\pi\xi \to 0$ for $0 \leq \xi \leq 1$ except $\xi = 0, \frac{1}{2}, 1$

$Z_n(\xi)$ converges to 0 almost surely.

7.42

$$\begin{aligned}
\xi &= \sum_{i=1}^{\infty} b_i 2^{-i} \\
P[b_1 = 0] &= P[b_1 = 1] = \frac{1}{2} \\
\therefore\ P[\xi < 0.5] &= P[\xi > 0.5] = \frac{1}{2} \\
P[b_1 b_2 = 00] &= P[b_1 b_2 = 01] = P[b_1 b_2 = .10] = P[b_1 b_2 = 11] = \frac{1}{4} \\
P[\xi < 0.25] &= P[0.25 < \xi < 0.5] = P[0.5 < \xi < 0.75] \\
&= P[\xi > 0.75] = \frac{1}{4}
\end{aligned}$$

and so on.

So ξ is uniformly distributed in [0,1].

b) If $b_1 = 0$ then we suppose a black ball was selected in first draw;
Thereafter we interpret $b_n = 1$ as "remove black ball, if any, from urn."

Similarly if $b_1 = 1$ then we suppose a white ball was selected in first draw;
Thereafter we interpret $b_n = 1$ as "remove white ball, if any, from urn."

7.43

5.43 a) $Y_n = 2^n X_1 X_2 ... X_n = \begin{cases} 2^n & X_1 = X_2 = ... = X_n = 1 \\ 0 & \text{otherwise} \end{cases}$

$\therefore Y_n \to 0$ almost surely.

b)
$$\begin{aligned} E[Y_n] &= 2^n P[X_1 = X_2 = ... = X_n = 1] \\ &\quad +0 \cdot (1 - P[X_1 = X_2 = ... = X_n]) \\ &= 2^n \left(\frac{1}{2}\right)^n \\ &= 1 \end{aligned}$$

Furthermore

$$E[Y_n^2] = (2^n)^2 \left(\frac{1}{2}\right)^n = 2^n \to \infty .$$

Thus Y_n does not converge to 0 in the m.s. sense.

7.44

$$\begin{aligned} E[(M_n - m)^2] &= E\left[\left(\frac{1}{n}\sum_i (X_i - m)\right)^2\right] \\ &= \frac{1}{n^2} E\left[\sum_i (X_i - m)^2\right] \\ &= \frac{1}{n^2} \cdot n\sigma^2 \end{aligned}$$

$$\therefore \quad \lim_{n\to\infty} E[(M_n - m)^2] = 0$$

7.45

~~5.45~~ We are given that $X_n \to X$ ms and $Y_n \to Y$ ms. Consider

$$\begin{aligned}\mathcal{E}[((X_n+Y_n)-(X+Y))^2] &= E[((X_n-X)+(Y_n-Y))^2] \\ &= E[(X_n-X)^2]+E[(Y_n-Y)^2] \\ &\quad +2E[(X_n-X)(Y_n-Y)]\end{aligned}$$

The first two terms approach zero since $X_n \to X$ and $Y_n \to Y$ in mean square sense. We need to show that the last term also goes to zero. This requires the Schwarz Inequality:

$$E[ZW] \le \sqrt{E[Z^2]}\sqrt{E[W^2]}\,.$$

When the inequality is applied to the third term we have:

$$\begin{aligned}E[((X_n+Y_n)-(X+Y))^2] &\le E[(X_n-X)^2]+E[(Y_n-Y)^2] \\ &\quad +2\sqrt{E[(X_n-X)^2]}\sqrt{E[(Y_n-Y)^2]} \\ &= (\sqrt{E[(X_n-X)^2]}+\sqrt{E[(Y_n-Y)^2]})^2 \\ &\to 0 \quad \text{as } n\to\infty\,.\end{aligned}$$

To prove the Schwarz Inequality we take

$$0 \le E[(Z+aW)^2]$$

and minimize with respect to a:

$$\begin{aligned}\frac{d}{da}(E[Z^2]+2aE[ZW]+a^2E[W^2]) &= 0 \\ 2E[ZW]+2aE[W^2] &= 0\end{aligned}$$

$\Rightarrow$ minimum attained by $a^* = -\dfrac{E[ZW]}{E[W^2]}$. Thus

$$0 \le E[(Z+a^*W)^2] = E[Z^2] = 2\frac{E[ZW]^2}{E[W^2]} + \frac{E[ZW]^2}{E[W^2]}$$

$$\begin{aligned}\Rightarrow \quad \frac{E[ZW]^2}{E[W^2]} &\le E[Z^2] \\ \Rightarrow \quad E[ZW] &\le \sqrt{E[Z^2]}\sqrt{E[W^2]} \quad \text{as required}\end{aligned}$$

7.46

5.46 a) No, X_n does not converge in m.s. sense.

b) X_n converges in distribution to $N(0,1)$.

7.47

7.42

5.47 Let ξ be as in Problem 5.42 and suppose the urn experiment is generated as in 5.42b. The first outcome b_1 is critical. If $b_1 = 0$ then

$$\begin{aligned} X_1(\xi) &= 1 \\ X_n(\xi) &= 1 \cdot b_1 \cdot b_2 \ldots b_n = \begin{cases} 1 & \text{if } b_2 = \ldots = b_n = 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Similarly if $b_1 = 1$ then $X_n(\xi) = 2$ all n. Now define

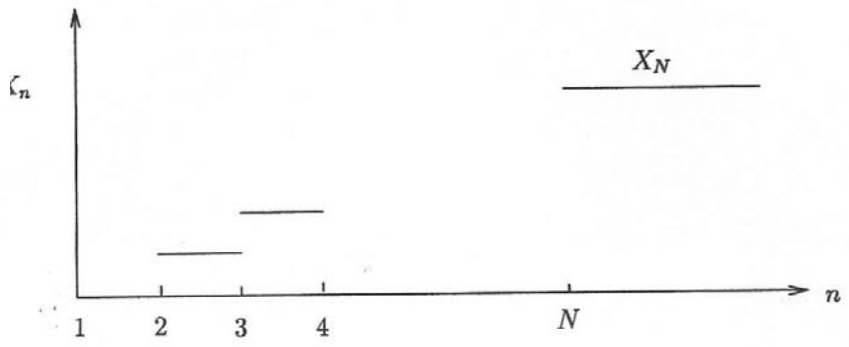
$$X(\xi) = \begin{cases} 0 & \text{if } b_1 = 0 \\ 2 & \text{if } b_1 = 1 \end{cases}$$

then

$$\begin{aligned} \mathcal{E}[(X_n(\xi) - X(\xi)^2] &= \mathcal{E}[(X_n(\xi) - 0)^2 | b_1 = 0] P[b_1 = 0] \\ &\quad + \mathcal{E}[(X_n(\xi) - 2)^2 | b_1 = 1] P[b_1 = 1] \\ &= \frac{1}{2} \mathcal{E}[X_n^2(\xi) | b_1 = 0] \\ &= \frac{1}{2} \cdot 1 \cdot P[b_2 = b_3 = \ldots = b_n = 1] \\ &= \left(\frac{1}{2}\right)^{n+1} \to 0 \,. \end{aligned}$$

7.48

a)



b) Let $A_1, A_2, \ldots$ be a sequence of iid Bernoulli trials with parameter p corresponding to potential customer arrivals. Let $B_1, B_2, \ldots$ be a sequence of iid Bernoulli trials with probability of success $\frac{99}{100}$, i.e. $P[B_i = 1] = \frac{99}{100} = \alpha$.

The sequence X_n is then given by

$$X_n = A_1B_1 + A_2B_1B_2 + \ldots + A_nB_1B_2 \ldots B_n + \ldots$$

Let N be the largest value of n for which $B_1 \ldots B_N = 1$, then

$$X_n = A_1 + \ldots + A_N \quad \text{for } n \geq N.$$

Now define the limiting random variable X by

$$X = A_1 + \ldots + A_N$$

where N is determined by the sequence of B_i's, then

$$X_n \to X \quad \text{as } n \to \infty \text{ almost surely.}$$

c)

$$\begin{aligned} \mathcal{E}[(X_n - X)^2] &= \mathcal{E}[\mathcal{E}[(X_n - X)^2|N]] \\ \mathcal{E}[(X_n - X)^2|N = k] &= \begin{cases} 0 & n \geq k \\ \mathcal{E}\left[\left(\sum_{i=n+1}^{k} A_i\right)^2\right] & n < k \end{cases} \end{aligned}$$

The summation above defines a Binomial random variable with parameters $k - n - 1$ and p, thus the second moment is:

$$\mathcal{E}\left[\left(\sum_{i=n+1}^{k} A_i\right)^2\right] = (k-n-1)pq + (k-n-1)^2p^2 \ .$$

Finally we have

$$\begin{aligned} \mathcal{E}[(X_n - X)^2] &= \sum_{k=1}^{\infty} \mathcal{E}[(X_n - X)^2|N = k]P[N = k] \\ &= \sum_{k=n+1}^{\infty} [(k-n-1)pq + (k-n-1)^2p^2]\alpha^k(1-\alpha) \\ &= \alpha^{n+1} \sum_{l=0}^{\infty} (lpq + l^2p^2)\alpha^l(1-\alpha) \\ &= \alpha^{n+1}\left[pq\frac{\alpha}{1-\alpha} + p^2\frac{\alpha+\alpha^2}{(1-\alpha)^2}\right] \\ &\to 0 \quad \text{as } n \to \infty \end{aligned}$$

Therefore X_n converges in mean square sense.

7.49

5.49 $Y_n = \cos 2\pi n\xi$. ξ: uniform RV in [0,1].

Note that there are $2n$ solutions to the equation

$$y = \cos 2\pi nx$$

for each fixed y in [-1,1]

$$\begin{aligned} f_{Y_n}(y_n) &= \sum_k \frac{f_X(x)}{|dy/dx|}\Big|_{x=x_k} \\ &= 2n \cdot \frac{1}{2\pi n\sqrt{1-y_n^2}} \\ &= \frac{1}{\pi\sqrt{1-y_n^2}} \end{aligned}$$

Define

$$\begin{aligned} F_Y(y) &= \int_{-1}^{y} \frac{1}{\pi\sqrt{1-y'^2}}dy' \\ F_n(y_n) &\to F_Y(y) \quad \text{as } n \to \infty. \end{aligned}$$

7.50

5.50 Laplacian RV

$$\begin{aligned} f_{X_n}(x_n) &= \frac{n}{2}\exp(-n|x_n|) \\ E[X_n] &= 0, \qquad VAR[X_n] = \frac{2}{n^2} \to 0 \quad \text{as } n \to \infty \end{aligned}$$

From Chebyshev inequality

$$P[|X_n - 0| \geq e] \leq \frac{\sigma^2}{e^2} \to 0 \quad \text{as } n \to \infty$$

The sequence converges in probability, and hence in distribution.

*7.5 Long-Term Arrival Rates and Associated Events

7.51

5.51 Let Y be the bus interdeparture time, then

$$Y = X_1 + X_2 + ... + X_m \quad \text{and} \quad \mathcal{E}[Y] = m\mathcal{E}[X_i] = mT$$

$$\therefore \text{ long-term bus departure rate} = \frac{1}{\mathcal{E}[Y]} = \frac{1}{mT}.$$

7.52

5.52 The time between forward ticks of the clock is a geometric random variable with $P[X = k] = (1-p)^{k-1}p$ and $\mathcal{E}[X] = \frac{1}{p}$.

$$\text{The long-term average tick rate} = \frac{1}{\mathcal{E}[X]} = p\frac{\text{sec}}{\text{sec}}.$$

7.53

5.53 Show $\{N(t) \geq n\} \Leftrightarrow \{S_n \leq t\}$.

a) We first show that $\{N(t) \geq n\} \Rightarrow \{S_n \leq t\}$.

$$\begin{aligned} \text{If } \{N(t) \geq n\} \Rightarrow t \geq S_{N(t)} &= X_1 + X_2 + \ldots + X_{N(t)} \\ &\geq X_1 + \ldots + X_n = S_n \\ \Rightarrow \{S_n \leq t\} & \end{aligned}$$

Next we show that $\{S_n \leq t\} \Rightarrow \{N(t) \geq n\}$. If $\{S_n \leq t\}$ then nth event occurs before time t

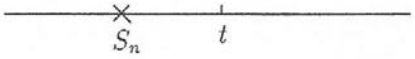
$$\begin{aligned} &\Rightarrow \quad N(t) \text{ is at least } n \\ &\Rightarrow \quad \{N(t) \geq n\} \quad \surd \end{aligned}$$

b)
$$\begin{aligned} P[N(t) \leq n] &= 1 - P[N(t) \geq n+1] \\ &= 1 - P[S_{n+1} \leq t] \\ &= 1 - \left(1 - \sum_{k=0}^{n} \frac{(\alpha t)^k}{k!} e^{-\alpha t}\right) \\ &= \sum_{k=0}^{n} \frac{(\alpha t)^k}{k!} e^{-\alpha t} \end{aligned}$$

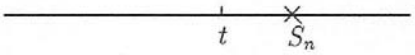
but S_{n+1} is an Erlang RV so by Ex. 3.16 we have that $N(t)$ is a Poisson random variable.

7.54

5.54 a) $\{N(t) = n\} \Rightarrow \{S_n \leq t\}$

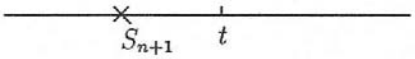


$\{S_n \geq t\}$

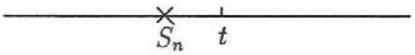


$\therefore \{N(t) = n\}$ and $\{S_n > t\}$ are mutually exclusive
$\therefore \{N(t) \leq n\}$ and $\{S_n \geq t\}$ cannot be equivalent

b) $\{N(t) > n\} \Rightarrow \{N(t) \geq n+1\} \Rightarrow \{S_{n+1} < t\}$



$\{S_n < t\}$



If $S_n < t < S_{n+1}$ then event $\{S_n > t\}$ occurs but event $\{N(t) > n\}$ doesn't; thus the two events cannot be equivalent.

7.55

5.55 Define

$$\begin{aligned} \text{cycle} &= \text{error free period} + \text{error period} \\ \text{"cost" per cycle} &= \text{duration of error free period} \end{aligned}$$

Then

$$\begin{aligned} \text{long-term proportion of time when channel is error free} &= \frac{\text{avg. cost for cycle}}{\text{avg. cycle length}} \\ &= \frac{m_1}{m_1 + m_2} \end{aligned}$$

7.56

5.56 $$\text{cycle} \triangleq \begin{matrix}\text{time boss}\\ \text{present}\end{matrix} + \begin{matrix}\text{time boss}\\ \text{absent}\end{matrix} = X_i + Y_i$$

$$\begin{aligned}\text{"cost"/cycle} &\triangleq \text{work done} = r_1 \times \begin{matrix}\text{time boss}\\ \text{present}\end{matrix} + r_2 \begin{matrix}\text{time boss}\\ \text{absent}\end{matrix}\\ &= r_1 X_i + r_2 Y_i\end{aligned}$$

$$\begin{aligned}\begin{matrix}\text{long term avg. rate at}\\ \text{which worker does work}\end{matrix} &= \frac{\mathcal{E}[r_1 X_i + r_2 Y_i]}{\mathcal{E}[X_i + Y_i]} = \frac{r_1 m_1 + r_2 m_2}{m_1 + m_2}\\ &= r_1 \frac{m_1}{m_1 + m_2} + r_2 \frac{m_2}{m_1 + m_2}\end{aligned}$$

7.57

5.57 a) $$\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mathcal{E}[X_1 + X_2 + X_3]} = \frac{1}{\mathcal{E}[X_1] + \mathcal{E}[X_2] + \mathcal{E}[X_3]}$$

b) Let $c_i(t) = \sum_{j=1}^{N(t)} X_{ij}$, $i = 1, 2, 3$. Then long term proportion of time spent servicing task i is

$$\lim_{t\to\infty} \frac{c_i(t)}{t} = \frac{\mathcal{E}[X_i]}{\mathcal{E}[X_1] + \mathcal{E}[X_2] + \mathcal{E}[X_3]}$$

c) Replace X_i by $Y_i = X_i + W_i$ then

$$\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mathcal{E}[X_1] + \mathcal{E}[X_2] + E[X_3] + 3\mathcal{E}[W]}$$

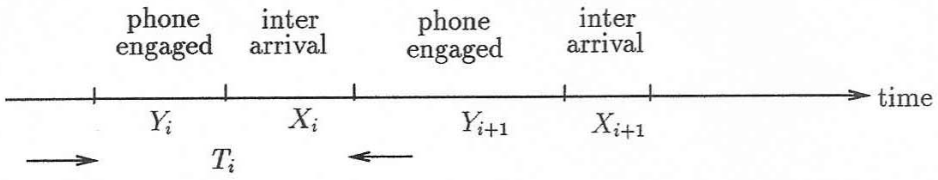
and

$$\lim_{t\to\infty} \frac{c_i(t)}{t} = \frac{\mathcal{E}[X_i]}{\mathcal{E}[X_1] + \mathcal{E}[X_2] + \mathcal{E}[X_3] + 3\mathcal{E}[W]}$$

7.58

~~5.58~~ a) Let T_i be the phone interseizure time, then

$$T_i = Y_i + X_i$$



$$\therefore \text{ long-term seizure rate} = \frac{1}{\mathcal{E}[T]} = \frac{1}{\mathcal{E}[Y] + \mathcal{E}[X]}$$

In order for the X_i to be iid, we need the memoryless property of the exponential RV.

b)

Define a "cycle"	=	# of customer arrivals in T_i
	=	$1 +$ # arrivals in $Y_i \triangleq 1 + N$
Define a "cost"	=	# customers that leave without using phone
	=	# arrivals in $Y_i \triangleq N$
Then long-term "cost" rate	=	long-term proportion of customers that leave without using phone
	=	$\frac{\mathcal{E}[N]}{1+\mathcal{E}[N]}$ From Ex. 4.26 $\mathcal{E}[N] = \lambda\mathcal{E}[Y]$
	=	$\frac{\lambda\mathcal{E}[Y]}{1+\lambda\mathcal{E}[Y]} = \frac{\mathcal{E}[Y]}{\frac{1}{\lambda}+\mathcal{E}[Y]} =$ proportion of time phone is engaged

7.59

5.59 The interreplacement time is

$$\tilde{X}_i = \begin{cases} X_i & \text{if } X_i < 3T \quad \text{that is item breaks down before } 3T \\ 3T & \text{if } X_i \geq 3T \quad \text{that is item is replaced at time } 3T \end{cases}$$

where the X_i are iid exponential random variables with mean $\mathcal{E}[X_i] = T$.

The mean of $\tilde{X}_i$ is:

$$\mathcal{E}[\tilde{X}_i] = \int_0^{3T} x\frac{1}{T}e^{-x/T}dx + 3TP[X > 3T] = T(1 - e^{-3})$$

a) Therefore the

$$\begin{matrix}\text{long-term} \\ \text{replacement} \\ \text{rate}\end{matrix} = \frac{1}{\mathcal{E}[\tilde{X}]} = \frac{1}{T(1 - e^{-3})}$$

b) Let

$$c_i = \begin{cases} 1 & X_i \geq 3T \\ 0 & X_i < 3T \end{cases} \quad \text{i.e. a good item is replaced}$$

Then

$$\mathcal{E}[C] = P[X_i \geq 3T] = e^{-3}$$

∴ long term rate at which working components are replaced is

$$\lim_{t\to\infty} \frac{\sum_{i=0}^{N(t)} C_i}{t} = \frac{\mathcal{E}[C]}{\mathcal{E}[\tilde{X}]} = \frac{e^{-3}}{T(1 - e^{-3})}$$

7.60

5.60 a) A codeword is produced each time a pattern is completed. The average pattern length X is

$$\mathcal{E}[X] = 1(0.1) + 2(0.9) + 3(0.81) + 4(0.0729 + .6561) = 3.439$$

$$\therefore \text{codeword production rate} = \frac{1}{3.439}\frac{\text{codewords}}{\text{ms}}$$

b)

$$\begin{aligned} \text{Define "cycle length"} &\triangleq \text{pattern length for one codeword} \\ \text{"cost"} &\triangleq \text{codeword length} \end{aligned}$$

$$\begin{matrix}\text{long term ratio} \\ \text{of encoded bits to} \\ \text{information bits}\end{matrix} = \frac{\mathcal{E}[C]}{\mathcal{E}[X]} = \frac{1(.6561)+3(1-.6561)}{3.439} = 0.49$$

7.61

5.61 a) $\mathcal{E}[X] = T \qquad F_X(y) = \frac{y}{2T} \qquad 0 < y < 2T$

$$\begin{matrix}\text{prop. of time} \\ r(t) \text{ exceed } c\end{matrix} = \frac{1}{T}\int_C^{2T}\left(1 - \frac{y}{2T}\right)dy = 1 - \frac{c}{T}\left(1 - \frac{c}{2T}\right)$$

b) $\mathcal{E}[X] = T \qquad F_X(y) = 1 - e^{-y/T}$

$$\begin{matrix}\text{prop. of time} \\ r(t) \text{ exceed } c\end{matrix} = \frac{1}{T}\int_C^{\infty}(1 - (1 - e^{-y/T}))dy = e^{-c/T}$$

c) $\mathcal{E}[X] = T \qquad F_X(y) = 1 - e^{-\pi y^2/4T^2}$

$$\begin{aligned}\begin{matrix}\text{prop. of time} \\ r(t) \text{ exceeds } c\end{matrix} &= \frac{1}{T}\int_c^{\infty}(1 - (1 - e^{-\pi y^2/4T^2}))dy \\ &= \frac{1}{T}\int_c^{\infty} e^{-\pi y^2/4^2}dy = 2Q\left(\sqrt{\frac{\pi}{2}}\frac{c}{T}\right)\end{aligned}$$

d) Let R be the residual time at a randomly-selected time instant. Then parts a)-c) involved finding $P[R > C] = 1 - P[R \leq C]$

X uniform:

$$\mathcal{E}[R] = \int_0^{\infty} P[R > x]dx = \int_0^{2T}\left(1 - \frac{x}{T} - \frac{x^2}{4T^2}\right)dx = \frac{2}{3}T$$

X exponential:

$$\mathcal{E}[R] = \int_0^{\infty} P[R > x]dx = \int_0^{2T} e^{-x/T}dt = T$$

X Rayleigh, the pdf of R is

$$\begin{aligned} f_R(x) &= -\frac{d}{dx}P[R > x] = \frac{1}{T}e^{-\pi x^2/4T^2} \\ \therefore \mathcal{E}[R] &= \int_0^{\infty} x f_R(x)dx = \int_0^{\infty}\frac{x}{T}e^{-\pi x^2/4T^2}dx = \frac{2}{\pi}T\end{aligned}$$

7.62

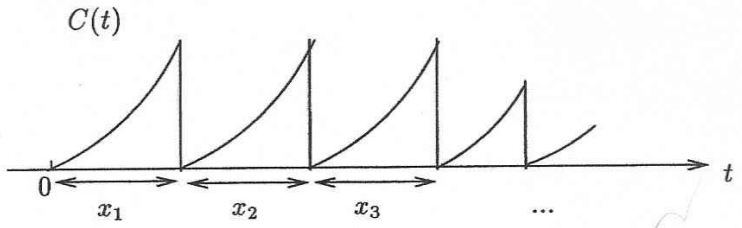
5.62 The amount of time that the age exceeds C in a cycle of length X is $(X - C)^+$. The long-term proportion of time that $a(t)$ exceeds C can be found by defining the cost per cycle by $C_j = (X_j - C)^+$. This is the same cost as considered in Example 5.23. Therefore the long-term proportion of time $a(t)$ exceeds C is given by Eqn. 5.50. 7.21 7.39

7.63

5.63 a) Since the age $a(t)$ is the time that has elapsed from the last arrival up to time t, then

$$C_j = \int_0^{X_j} a(t')dt' = \int_0^{X_j} t'dt' = \frac{X_j^2}{2}$$

The figure below shows the relation between $a(t)$ and the C_j's.



b) $$\lim_{t\to\infty} \frac{C(t)}{t} = \frac{\mathcal{E}[C]}{\mathcal{E}[X]} = \frac{\mathcal{E}[X^2]}{2\mathcal{E}[X]}$$

c) From the above figure:

$$\begin{aligned}
\lim_{t\to\infty} \frac{1}{t}\int_0^t a(t')dt' &= \lim_{t\to\infty} \frac{1}{t}\sum_{j=1}^{N(t)} \int_0^{X_j} a(t')dt' \\
&= \lim_{t\to\infty} \frac{1}{t}\sum_{j=1}^{N(t)} C_j \\
&= \frac{\mathcal{E}[X^2]}{2\mathcal{E}[X]} \quad \text{from part b)}
\end{aligned}$$

d) For the residual life in a cycle

$$C_j' = \int_0^{X_j} r(t')dt' = \int_0^{X_j} (X_j - t')dt' = \frac{X_j^2}{2} = C_j$$

$\Rightarrow$ same cost as for age of a cycle

7.64

From Prob. ~~5.63~~ 7.63

$$\mathcal{E}[R] = \frac{\mathcal{E}[X^2]}{2\mathcal{E}[X]}$$

X uniform is $(0, 2T)$

$$\mathcal{E}[X^2] = \frac{1}{2T}\int_0^{2T} t^2 dt = \frac{1}{2T}\frac{(2T)^3}{3} = \frac{4T^2}{3}$$
$$\mathcal{E}[R] = \frac{4T^2}{2(3)T} = \frac{2T}{3}$$

X exp. with mean T

$$\mathcal{E}[X^2] = VAR[X] + \mathcal{E}[X]^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = 2T^2$$
$$\mathcal{E}[R] = \frac{2Y^2}{2T} = T$$

X Rayleigh

$$\mathcal{E}[X^2] = VAR(X) + \mathcal{E}[X]^2 \quad \mathcal{E}[X] = \alpha\sqrt{\frac{\pi}{2}} = T$$
$$= \left(2 - \tfrac{\pi}{2}\right)\alpha^2 + \alpha^2\tfrac{\pi}{2} \quad \Rightarrow \alpha^2 = T^2\frac{2}{\pi}$$
$$= 2\alpha^2$$
$$= \tfrac{4T^2}{\pi}$$
$$\mathcal{E}[R] = \frac{4T^2}{2\pi T} = \tfrac{2T}{\pi}$$

7.65

5.65 a) Define the length of a cycle by N_j and the cost of a cycle by T_j, then

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{\infty}D_k=\lim_{n\to\infty}\frac{\sum_{j=1}^{\infty}T_j}{\sum_{j=1}^{\infty}N_j}=\frac{\mathcal{E}[T]}{\mathcal{E}[N]}$$

b) Let the simulation run for exactly k regeneration cycles and let

$$\begin{aligned} T_j &= \text{total delay during } j\text{th cycle} \\ N_j &= \#\text{ customers served during } j\text{th cycle} \end{aligned}$$

then (T_j, N_j) are an iid sequence and

$$< T >= \frac{1}{k}\sum_{j=1}^{k}T_j \qquad < N >= \frac{1}{k}\sum_{j=1}^{k}N_j$$

are unbiased estimates for $\mathcal{E}[T]$ and $\mathcal{E}[N]$.

Estimate mean delay by $\dfrac{< T >}{< N >}$.

*7.6 Calculating Distributions Using the Discrete Fourier Transform

7.66

5.66 a)

$$\begin{aligned} c_0 &= \frac{1}{3}+\frac{1}{3}+\frac{1}{3}=1 \\ c_1 &= \frac{1}{3}+\frac{1}{3}e^{j\frac{2\pi}{3}}+\frac{1}{3}e^{j\frac{2\pi}{3}}=0 \\ c_2 &= \frac{1}{3}+\frac{1}{3}e^{j\frac{4\pi}{3}}+\frac{1}{3}e^{j\frac{8\pi}{3}}=0 \end{aligned}$$

b) $P[X=1]=\frac{1}{3}\left[1+0\cdot e^{-j\frac{2}{3}\pi}+0\cdot e^{-j\frac{4}{3}\pi}\right]=\frac{1}{3}$

7.67

$$\Phi_Z(\omega) = \Phi_X(\omega)^2\left(\frac{1}{3}+\frac{1}{3}e^{j\omega}+\frac{1}{3}e^{j2\omega}\right)^2$$

$$c_m = \Phi_Z\left(\frac{2\pi m}{5}\right)=\left(\frac{1}{3}+\frac{1}{3}e^{j\frac{2\pi m}{5}}+\frac{1}{3}e^{j\frac{4\pi m}{5}}\right)^2 \qquad m=0,\ldots,4$$

a)

$$\begin{aligned} \Rightarrow c_0 &= 1 \\ c_1 &= \frac{1}{9}(1+e^{j2\pi/5}+e^{j4\pi/5})^2=-0.235+j(0.171) \\ c_2 &= 0.0131+j(-0.04) \\ c_3 &= 0.0131+j(0.04) \\ c_4 &= -0.235-j(0.171) \end{aligned}$$

b)

$$\begin{aligned} P[S=2] &= \frac{1}{5}\sum_{m=0}^{4} c_m e^{-j4\pi m/5} \\ &= \frac{1}{5}[1+(-.235+j(.171))e^{-j4\pi/5} \\ &\quad +(.0131-j(.04))e^{-j2\pi/5} \\ &\quad +(.0131+j(.04))e^{-j12\pi/5} \\ &\quad +(-.235-j(.171))e^{-j16\pi/5}] \\ &= \frac{1}{(1.666)}=\frac{1}{3} \end{aligned}$$

7.68 The following Octave code produces the FFTs:

```
N = 8;
P = 1/2;
n = [0:N-1];
cms = fft(binomial_pdf(n, N, P), 16);
%You can also evaluate the characteristic function directly...
%w = 2.*pi.*n./N;
%cms = (1-P+P.*e.^(j.*w)).^N;
pmf = ifft(cms.*cms);
figure;
stem([1:16], pmf, "b");
```

7.69 The following Octave code produces the FFTs:

```
N = 5;
P = ones(1,10)./10;
pmf = ifft(fft(P, 9.*N).^N);
stem([0:9.*N-1], pmf);

N = 10;
P = ones(1,10)./10;
pmf = ifft(fft(P, 9.*N).^N);
stem([0:9.*N-1], pmf);
```

7.70 The following Octave code produces the FFT for evaluating Eq. (7.55):

```
N = 8;
P = 1/2;
n = [0:N-1];
w = 2.*pi.*n./N;
cms = P.*e.^(j.*w)./(1-(1-P).*e.^(j.*w));
pmf = ifft(cms);
figure;
stem([N-1:-1:0], pmf);
ek = (1-P).*P.^n.*(P.^N./(1-P.^N));

N = 16;
P = 1/2;
n = [0:N-1];
w = 2.*pi.*n./N;
cms = P.*e.^(j.*w)./(1-(1-P).*e.^(j.*w));
pmf = ifft(cms);
figure;
stem([N-1:-1:0], pmf);
ek = (1-P).*P.^n.*(P.^N./(1-P.^N));
```

7.71 The following Octave code produces the FFTs:

```
%N=16 achieves an error percent less than 0.01. This can be found
%by simple trial and error using different values for N.
N = 16;
L = 5;
n = [0:N-1];
w = 2.*pi.*n./N;
cms = e.^(L.*(e.^(j.*w)-1));
pmf = ifft(cms);
figure;
stem([N-1:-1:0], pmf);

N = 5;
L = 5;
n = [0:19];
P = poisson_pdf(n, L);
pmf = ifft(fft(P, 65).^N);
stem([0:64], pmf);
```

7.74 The following Octave code produces the FFT to obtain the pdf of Z:

```
function fx = ift(phix, n, N)
    phixs = [phix((N/2+1):N) phix(1:(N/2))];
    fxs = fft(phixs)./(2.*pi);
    fx = fftshift(fxs);
end

N = 512;
n = [-(N/2):(N/2-1)];
d = 2.*pi.*n./N;
alphaX = 1;
alphaY = 2;
phiX = 1./(1 + alphaX.^2.*n.^2);
phiY = 1./(1 + alphaY.^2.*n.^2);
phiZ = phiX.*phiY;
pdf = ift(phiZ, n, N);
figure;
plot(d, pdf, "b");
hold on;
plot(d, ift(phiX, n, N), "g");
plot(d, ift(phiY, n, N), "r");
```

7.75 The following Octave code produces the FFT to obtain the pdf of a Gaussian random variable:

```
function fx = ift(phix, n, N)
    phixs = [phix((N/2+1):N) phix(1:(N/2))];
    fxs = fft(phixs)./(2.*pi);
    fx = fftshift(fxs);
end

N = 512;
n = [-(N/2):(N/2-1)];
d = 2.*pi.*n./N;
phiX = e.^(-n.^2./2);
pdf = ift(phiX, n, N);
figure;
plot(d, pdf, "b");
hold on;
plot(d, normal_pdf(d, 0, 1), "r");
```

7.76 The following Octave code produces Figs. 7.2 through 7.4:

```
function stepplot(x, y)
    xn = zeros(1, 2*length(x));
    yn = zeros(1, 2*length(x));
    for k = 1:length(x)
        xn(2*k-1) = x(k);
        xn(2*k) = x(k) + 1;
        yn(2*k-1) = y(k);
        yn(2*k) = y(k);
    end
    plot(xn, yn);
end

%Figure 7.2a
N = 5;
P = ones(1,2)./2;
pmf = ifft(fft(P, N+1).^N);
cdf = zeros(size(pmf));
for i = 1:N+1
    cdf(i) = sum(pmf(1:i));
end
figure;
stepplot([0:N], cdf);
hold on;
x = [0:0.01:N+1];
plot(x, normal_cdf(x, 2.5, 1.25), "r");

%Figure 7.2b
N = 25;
P = ones(1,2)./2;
pmf = ifft(fft(P, N+1).^N);
cdf = zeros(size(pmf));
for i = 1:N+1
    cdf(i) = sum(pmf(1:i));
end
figure;
stepplot([0:N], cdf);
hold on;
x = [0:0.01:N+1];
plot(x, normal_cdf(x, 12.5, 6.25), "r");

%Figure 7.3
N = 5;
P = ones(1,10)./10;
pmf = ifft(fft(P, 9.*N+1).^N);
cdf = zeros(size(pmf));
for k = 1:9.*N+1
    cdf(k) = sum(pmf(1:k));
end
figure;
stepplot([0:9.*N], cdf);
hold on;
x = [0:0.01:9.*N+1];
plot(x, normal_cdf(x, 22.5, 41.3), "r");
```

Problems Requiring Cumulative Knowledge

7.77

a) $S = X + Y$

Assume X and Y are independent

$$\begin{aligned} G_S(z) &= G_X(z)G_Y(z) \\ &= (pz + 1 - p)^n(rz + 1 - r)^m \end{aligned}$$

b) $G_S(z) = E[z^S] = \sum_{k=0}^{n+m} z^k P[S = k]$

$P[S = k]$ is the coefficient of z^k in $G_S(z)$.

$$\begin{aligned} G_S(z) &= (pj + 1 - p)^n(rj + 1 - r)^m \\ P[S = k] &= \sum_{j=0}^{k} \binom{n}{j} p^j(1-p)^{n-j} \binom{m}{k-j} r^{k-j}(1-r)^{m-(k-j)} \end{aligned}$$

c) Obtain $G_S(z)$ at $z = \exp(j2\pi k/(n+m))$. Use IDFT to obtain p_k.

7.78

~~5.78~~ $X_n = \frac{1}{2}U_n + \left(\frac{1}{2}\right)^2 U_{n-1} + \ldots + \left(\frac{1}{2}\right)^n U_1 \quad n \geq 1$

This "low-pass filter" weighs recent samples more heavily than older samples. Note that we can also write X_n as follows:

$$X_n = \frac{1}{2}X_{n-1} + \frac{1}{2}U_n \qquad X_0 = 0, \quad n \geq 1$$

We will see in Chapter 6 that X_n is an autoregressive random process.

a)

$$\begin{aligned}
E[X_n] &= E\left[\frac{1}{2}\sum_{j=0}^{n-1}\left(\frac{1}{2}\right)^j U_{n-j}\right] = \frac{1}{2}\sum_{j=0}^{n-1}\left(\frac{1}{2}\right)^j E[U] \\
&= \frac{1}{2}E[U]\frac{1-(\frac{1}{2})^n}{1-\frac{1}{2}} = E[U]\left(1-\left(\frac{1}{2}\right)^n\right) \\
&= 0 \quad \text{since } E[U] = 0. \\
E[X_n^2] &= E\left[\frac{1}{2}\sum_{j=0}^{n-1}\left(\frac{1}{2}\right)^n U_{n-j}\ \frac{1}{2}\sum_{j'=0}^{n-1}\left(\frac{1}{2}\right)^{j'} U_{n-j'}\right] \\
&= \frac{1}{4}\sum_{j=0}^{n-1}\sum_{j'=0}^{n-1}\left(\frac{1}{2}\right)^{j+j'} E[U_{n-j}U_{n-j'}] \\
&= \frac{1}{4}\sum_{j=0}^{n-1}\left(\frac{1}{2}\right)^{2j} E[U^2] \quad \text{since the } U_j \text{ are iid} \\
&= \frac{\sigma^2}{3}\left(1-\left(\frac{1}{4}\right)^n\right) \quad \text{where } E[U^2] = \sigma^2 \\
VAR(X_n) &= E[X_n^2] - E[X_n]^2 = \frac{\sigma^2}{3}\left(1-\left(\frac{1}{4}\right)^n\right).
\end{aligned}$$

Thus X_n is a zero-mean Gaussian random variable with the variance found in part a).

As $n \to \infty$

$$\Phi_{X_n}(\omega) \to e^{-\frac{1}{2}\frac{\sigma^2}{3}\omega^2}$$

so X_n approaches a zero-mean Gaussian random variable with variance $\sigma^2/3$.

c) The result in Part b) shows that X_n converges in distribution to a Gaussian random variable X with zero mean and variance $\sigma^2/3$.

To determine whether X_n converges in mean-square sense consider the Cauchy Criterion in Eq. 5.50. Consider X_n and X_{n+m}:

$$\begin{aligned}
\mathcal{E}[(X_{n+m} - X_n)^2] &= \mathcal{E}\left[\left(\frac{1}{2}\sum_{j=0}^{n+m-1}\left(\frac{1}{2}\right)^n U_{n-j} - \frac{1}{2}\sum_{j'=0}^{n+m-1}\left(\frac{1}{2}\right)^{j'} U_{n-j'}\right)^2\right] \\
&= \frac{1}{4}\mathcal{E}\left[\left(\sum_{j=n}^{n+m-1}\left(\frac{1}{2}\right)^j U_{n-j}\right)^2\right] \\
&= \frac{1}{4}\sum_{j=n}^{n+m-1}\sum_{j'=n}^{n+m-1}\left(\frac{1}{2}\right)^{j+j'} E[U_{n-j}U_{n-j'}] \\
&= \frac{\sigma^2}{4}\sum_{j=n}^{n+m-1}\left(\frac{1}{2}\right)^{2j} \\
&= \frac{\sigma^2}{4}\left(\frac{1}{4}\right)^n\left(\frac{1-(\frac{1}{4})^m}{1-\frac{1}{4}}\right) \\
&\to 0 \quad \text{as } n, m \to \infty
\end{aligned}$$

Therefore X_n converges in mean-square sense.

To determine almost-sure convergence of X_n would take us beyond the scope of the text. See Gray and Davisson, page 183 for a discussion on how this is done.

7.79

a) $S_n = X_1 + X_2 + ... + X_n$, S_n is Gaussian

$$\begin{aligned} E[S_n] &= n\mu \\ VAR[S_n] &= E[[(X_1 - m) + ... + (X_n - m)]^2] \\ &= (\sigma^2 + p\sigma^2) + (p\sigma^2 + \sigma^2 + p\sigma^2) + (p\sigma^2 + \sigma^2 + p\sigma^2) \\ &\quad +... + (\sigma^2 + \sigma^2 + p\sigma^2) + (p\sigma^2 + \sigma^2) \\ &= n\sigma^2 + [1 + 2(n-2) + 1]p\sigma^2 \\ &= (n + (2n-2)p)\sigma^2 \\ \phi_{S_n}(\omega) &= e^{jn\mu\omega - [n+2np-2p]\sigma^2\omega^2/2} \end{aligned}$$

b) Suppose $n \geq m$.

$$\begin{aligned} S_n - S_m &= X_{m+1} + X_{m+2} + ... + X_n \quad \text{also Gaussian} \\ E[S_n - S_m] &= (n-m)\mu \\ VAR[S_n - S_m] &= [(n-m) + 2(n-m)p - 2p]\sigma^2 \end{aligned}$$

c) Assume $n < m$.

$$\begin{aligned} \phi_{S_m,S_n}(\omega_1,\omega_2) &= E[e^{j\omega_1 S_m + j\omega_2 S_n}] \\ &= E[E[e^{j\omega_1 S_m + j\omega_2 S_n}|S_m]] \\ &= E[e^{j\omega_1 S_m} E[e^{j\omega_2 S_n}|S_m]] \\ &= E\left[e^{j\omega_1 S_m}\left\{\exp\left[j\omega_2 n\mu - \frac{\omega_2^2\sigma^2}{2}[(n-m+2(n-m)p-2p]\right]\right\}\right] \\ &= \exp\left\{j\omega_1 m\mu - \frac{\omega_1^2\sigma^2}{2}[m + (2m-2)p]\right| \cdot \\ &\quad \exp\left\{j\omega_2 n\mu - \frac{\omega_2^2\sigma^2}{2}[(n-m) + 2(n-m)p - 2p]\right\}\} \end{aligned}$$

d) No.

7.80)

$$S_n = \sum_{i=1}^{n} X_i$$

a) $$E[S_n] = \sum_{i=1}^{n} E[X_i] = n\mu$$

$$VAR[S_n] = E\left[\left[(X_1-\mu_1)+\cdots+(X_n-\mu_n)\right]^2\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} COV(X_i, X_j)$$

$$= n\sigma^2 + (\rho\sigma^2) + (\rho\sigma^2+\rho\sigma^2) + \cdots + (\rho\sigma^2+\rho\sigma^2) + \rho\sigma^2$$

$$= n\sigma^2 + 2\rho\sigma^2 + 2(n-2)\rho\sigma^2 = n\sigma^2 + (2n-2)\rho\sigma^2 = (n+(2n-2)\rho)\sigma^2$$

S is a Gaussian RV, Therefore:

$$\Phi_{S_n}(\omega) = e^{\left\{j\omega n\mu\right) - \frac{1}{2}(n+(2n-2)\rho)\sigma^2\omega^2\}}$$

b) Suppose $n > m$

$S_n - S_m = X_{m+1} + \cdots + X_n$ is also Gaussian

$$E[S_n - S_m] = E[S_n] - E[S_m] = (n-m)\mu$$

$$VAR[S_n - S_m] = \sum_{i=m+1}^{n}\sum_{j=m+1}^{n} COV(X_i, X_j) = [(n-m) + 2(n-m)\rho - 2\rho]\sigma^2$$

P7.80)

c) S_m & $S_n - S_m$ are independent if $m < n$

we have: $\varphi_{S_n - S_m}(\omega) = e^{\{ j\omega \quad / - \frac{\sigma^2\omega^2}{2}((n-m) + 2(n-m)\rho - 2\rho] \}}$

$\varphi_{S_m}(\omega) = e^{\{ j\omega \quad) - \frac{1}{2}(m + (2m-2)\rho)\sigma^2\omega^2 \}}$

Therefore: $\varphi_{S_m, S_n}(\omega_1, \omega_2) = E[e^{j\omega_1 S_m + j\omega_2 S_n}]$

$= E\left[E\left[e^{j\omega_1 S_m + j\omega_2 S_n} | S_m\right]\right]$

$= E\left[e^{j\omega_1 S_m} E\left[e^{j\omega_2 S_n} | S_m\right]\right]$

$= E\left[e^{j\omega_1 S_m}\right] E\left[e^{j\omega_2 S_n} | S_m\right]$

$= E\left[e^{j\omega_1 S_m}\right] E\left[e^{j\omega_2 (S_n - S_m)}\right]$

$= \varphi_{S_m}(\omega_1)\, \varphi_{S_n - S_m}(\omega_2)$

d) using Cauchy Criterion, it can be seen that it doesn't converge.

7.81

~~5.80~~ Z_n does not converge in mean square sense.

7.82

~~5.81~~ **a)**

$$\begin{aligned} E[Y_n] &= -\Sigma P[X_n]\log_2 P[X_n] = H(X) \\ &= \frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{16}\cdot 4 + \frac{1}{16}\cdot 4 \\ &= 1\frac{7}{8} \\ VAR[Y_n] &= E[(Y_n - 1\frac{7}{8})^2] \\ &= \frac{1}{2}(1-1\frac{7}{8})^2 + \frac{1}{4}(2-1\frac{7}{8})^2 + \frac{1}{8}(3-1\frac{7}{8})^2 + 2\frac{1}{16}(4+1\frac{7}{8})^2 \\ &= 1.11 \end{aligned}$$

b) Functions of independent RVs are also independent, so $\{Y_n\}$ is an iid sequence. The weak law and strong of large numbers apply.

c) The long term bit rate $S_n = \frac{1}{n}\sum_{k=1}^{n} L(x_k) = \frac{1}{n}\sum_{k=1}^{n} Y_k$ will approach the entropy H_x of the source with probability 1.